Simple Linear Regression

2)Implement a linear regression algorithm to predict a continuous target variable based on one predictor variables. E.g. Implement a linear regression model on a housing prices dataset to predict the prices of houses based on anyone features

Ans) `import pandas as pd`

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Dataset: Hours studied and marks obtained
data = {
    'Hours_Studied': [1, 2, 3, 4, 5, 6, 7, 8],
    'Marks': [35, 42, 50, 55, 62, 70, 78, 85]
}

# Convert data into DataFrame
dataset = pd.DataFrame(data)

# One predictor variable
X = dataset[['Hours_Studied']]

# Target variable
y = dataset['Marks']

# Create the model
regression_model = LinearRegression()

# Train the model
regression_model.fit(X, y)

# Predict marks
predicted_marks = regression_model.predict(X)

# Display the regression equation values
print("Slope:", regression_model.coef_[0])
print("Intercept:", regression_model.intercept_)
```

```
# Predict marks for a student studying 5.5 hours
new_hours = pd.DataFrame({'Hours_Studied': [5.5]})
prediction = regression_model.predict(new_hours)

print("Predicted Marks for 5.5 hours:", prediction[0])

# Plot actual data
plt.scatter(X, y, label="Actual Data")

# Plot regression line
plt.plot(X, predicted_marks, label="Regression Line")

plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Simple Linear Regression")
plt.legend()
plt.show()
```

```
Slope: 7.107142857142858
Intercept: 27.64285714285714
Predicted Marks for 5.5 hours: 66.73214285714286
```

